

Sadegh Mehrabi

Embedded System Engineer

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E-mail: s.mehrabi@aut.ac.ir Cell phone: (+98) 917-145-6706 [Website](#) [Linkedin Profile](#)

Embedded System Engineer with 8+ years of experience in **Firmware Development, Embedded C/C++, RTOS, Bare-Metal Programming** for microcontroller-based systems. Experienced in hardware-software integration, communication protocols, and real-time data processing. Familiar with sensor-based systems, system validation, and cross-domain development including firmware, hardware interfacing, and tooling. Uses structured coding styles including **CamelCase** and **snake_case** naming conventions depending on project requirements, with consistent C/C++ code formatting, **modular design principles**, and **MISRA C/C++** guidelines for **safety-critical** embedded software.

Skills

- **Embedded & Real-Time Systems**
 - Embedded C/C++ development for ARM Cortex-M/R, DSP, and AVR microcontrollers.
 - RTOS and bare-metal firmware development.
 - CAN, LIN, SPI, I2C, UART, Ethernet TCP/IP, Wi-Fi, Bluetooth, and ARINC 429.
 - Bootloaders, DMA, interrupts, timers, queues, and memory management
- **Software Development**
 - HTML, CSS, JavaScript, C#, .NET desktop application development
 - Qt/Qt Quick and Unity-based tools.
 - MVVM, MVC, and SOLID design principles.
 - Git, GitLab, GitHub, CI/CD, and automated build systems.
- **Power Electronics**
 - Power electronics fundamentals and converter topologies (Buck, Boost, Flyback, Full-Bridge).
 - Design of isolated AC-DC (3.3 kW, 400 V) and DC-DC (2.4 kW, 12 V) converters.
 - Protection systems design including OVP, UVP, OCP, OTP, inrush current limiting (ICL), short-circuit protection, and EMI filters.
 - Simulation and validation using MATLAB/Simulink, PSIM, and PSpice.
- **Electronic Hardware Design**
 - Schematic and multilayer PCB design using Altium Designer.
 - EMI/EMC considerations, impedance control, and signal integrity.
- **Control, Signal Processing & Robotics (Background)**
 - PID, LQR, and Fuzzy control algorithms.
 - DC and stepper motor control (position, speed, and torque control).
 - Sensor fusion and filtering algorithms including EKF, UKF, and complementary filters.
- **Engineering Tools**
 - MATLAB/Simulink, PSIM, PSpice, Proteus, Altium Designer.
 - Visual Studio, Qt Creator, Unity, STM32CubeIDE, Code Composer Studio, Keil, IAR.
- **Soft Skills**
 - Problem solving, Clear technical communication, Attention to detail, Independent working ability, and Time management and prioritization of tasks

Experience

May.2023–Present
Full time

Firmware & Software developer
Luminocity AG , Switzerland / Remote

- Developed real-time data acquisition and processing systems for multi-sensor devices (GNSS, IMU, magnetometer, barometer, and camera) using **FreeRTOS**.

- Developed modular embedded software architecture focusing on performance, scalability, and maintainability.
- Built desktop applications for real-time monitoring and visualization of sensor data using **.NET MAUI Blazor**.
- Developed **C#** and **C++** components for system integration and data processing.
- Contributed to **CI/CD** workflows using **GitLab** and **GitHub** for automated build, testing, and deployment processes.
- Implemented unit and integration testing using **xUnit** and **Nunit**.

Sep.2020–Apr. 2023
Full time

Software developer & Hardware designer

PishroEdehAraz Co. Tehran, Iran

- Developed embedded firmware using **FreeRTOS** for real-time control and communication (**ARM Cortex-M7**), considering **functional safety** requirements (**ISO 26262**).
- Designed **schematics** and **PCB layouts** for embedded and power electronics systems with focus on EMI and reliability.
- Implemented **Modbus TCP/RTU** communication for industrial control systems (**ARINC 429** converters, throttle control).
- Designed and analyzed **AC-DC** and **DC-DC** converters for EV applications.
- Designed isolated power (3.3 kW AC-DC, 2.4 kW DC-DC) using TI **C2000**.
- Implemented protection features (OVP, UVP, OCP, OTP, ICL, EMI filtering).
- Performed simulation and validation using **MATLAB/Simulink**, **PSIM**, and **PSpice**.
- Worked with power converter topologies (**Buck**, **Boost**, **Flyback**, **Full-Bridge**).
- Developed communication including **RS485** and Ethernet **TCP/IP (LWIP)**.

Jun.2018– Sep. 2020
Part time

Firmware developer & Hardware designer

AradSharif Co. Tehran, Iran

- Implemented Ethernet communication (**TCP/IP**, **UDP**, **ICMP**) on **STM32** using **LWIP** and **RL-TCPnet** and **RTOS**.
- Developed secure **bootloader** with firmware encryption for safe **OTA** firmware updates.
- Designed schematics and multilayer PCBs for embedded systems.

Apr.2016– Feb.2020
Part time

Software developer & Hardware designer

Kharazmi Co.(Redronic) Tehran, Iran

- Developed low cost **AHRS** (Attitude and Heading Reference System) embedded solution with accuracy of 0.1° in **roll**, **pitch**, and 0.5° in **yaw**.
- Implemented **magnetometer calibration** and magnetic disturbance detection algorithms.
- Implemented sensor data fusion algorithms including **Kalman filter** (KF), EKF, MAH, and MAD.
- Processed GPS data for INS integration and navigation applications.
- Designed **INS** hardware and contributed to system integration.
- Designed and controlled a **6-RSS parallel robot** as a laboratory project.
- Used hardware-in-the-loop (**HIL**) and software-in-the-loop (**SIL**) with MATLAB for validation.
- Designed, modeled, and controlled a **self-balancing robot** and validated **PID** and digital **LQR** controllers.

Education

- **Master of Science (MSc), Electrical Engineering (Control)**

Sep. 2015 – Jan. 2018

- Amirkabir University of Technology, Tehran, Iran.
- GPA: **17.7/20** (32 credits) · Ranked **4th** in Electrical Engineering

- **Bachelor of Science (BSc), Electrical Engineering (Control)**

Sep. 2010 – Sep. 2014

- Qom University of Technology, Qom, Iran.
- GPA: **17.95/20** (141 credits) · Ranked **1st** in Electrical Engineering

❖ **Mobile mapping system (2025)**

Developed a high-precision real-time aerial mapping and navigation system based on dual-core **STM32H7** microcontrollers. Implemented embedded firmware using **FreeRTOS** with multi-tasking real-time architecture for synchronized sensor processing. Integrated **IMU sensors (accelerometer, gyroscope, magnetometer, barometer)** and **dual GNSS** modules for positioning and orientation estimation in aerial mapping applications.

Implemented interrupt-driven high-speed data acquisition and synchronized sensor logging to **SD card**, with **external flash** memory management for configuration and system parameters. Designed and integrated communication interfaces including **Bluetooth, Wi-Fi, and Ethernet** for data transfer and connectivity.

Developed **multi-camera** synchronization and triggering system supporting up to **6 cameras** for aerial photogrammetry applications.

Developed a desktop monitoring and visualization application using **C# and .NET (Blazor), JavaScript, HTML/CSS, and SQLite** for real-time sensor data analysis and system monitoring.

Contributed to system architecture, hardware integration, debugging, and coordination of embedded hardware development tasks with focus on real-time performance and system reliability.

❖ **Medical Auto Sampler system aligned with IEC 62304 (2024)**

Developed an automated medical autosampler system for blood sample processing in laboratory environments. Designed embedded hardware and implemented firmware in **Embedded C++ on STM32F4** microcontrollers using **FreeRTOS**.

Designed and implemented a fully automated sample handling system supporting up to 30 blood sample tubes for laboratory workflows.

Integrated a **touchscreen** user interface and **USB HID** interface for operator control and system configuration. Implemented **RS232** communication with **ProLYTE** medical analyzers for command exchange, synchronization, and data acquisition.

Developed data management functionality for storing up to 300 patient records and test results.

Focused on real-time task management, hardware-software integration, and reliable serial communication for stable laboratory operation.

❖ **Arinc429 converter system (2023)**

Developed a high-speed **ARINC429 to Modbus TCP** protocol converter for aerospace applications using **Embedded C++ on STM32H7** microcontrollers. Designed system hardware architecture, schematics, and embedded firmware for real-time avionics communication.

Implemented support for 32 ARINC429 receive channels and 16 transmit channels for multi-channel communication with aircraft subsystems.

Developed bidirectional data conversion between ARINC429 and Modbus TCP over **100 Mbps Ethernet**, enabling real-time data exchange with external monitoring and control systems.

Implemented firmware using **FreeRTOS** with multiple concurrent tasks, focusing on deterministic execution, low-latency communication, and reliable data handling.

❖ **On board converter system for Electric Vehicle (2022)**

Developed an automotive-grade EV charger with **single-phase (3.3 kW)** and **three-phase (10 kW)** operation. Designed power electronics hardware including schematics, control-stage architecture, magnetic component design, simulation, and **8-layer PCB** layout with focus on **EMC/EMI/ESD** considerations (**ISO 7637, ISO 16750**) and **functional safety principles aligned with ISO 26262**.

Implemented embedded control firmware in **Embedded C++ on TI C2000** microcontrollers using a bare-metal architecture for real-time control. Developed and tuned digital control algorithms including **Adaptive PID** for power regulation, using topologies such as **Phase-Shifted Full-Bridge (PSFB)** and **Interleaved PFC** for efficiency and thermal performance optimization.

Integrated **CAN 2.0** communication (**ISO 11898**) for vehicle-charger interaction and system monitoring.

Developed a **Qt-based desktop** application for testing, debugging, and system validation via serial-to-CAN interface.

❖ **Attitude and Heading Reference System (AHRS) (2021)**

Developed a high-precision **AHRS** (Attitude and Heading Reference System) for real-time orientation and motion estimation. Designed embedded hardware and implemented firmware in **Embedded C** using **bare-metal programming** for sensor acquisition, signal processing, and attitude estimation.

Implemented sensor fusion algorithms using gyroscope, accelerometer, and magnetometer data to compute **Euler angles, quaternions, and angular velocity**.

Developed filtering and estimation methods including **Kalman Filter-based** approaches for orientation estimation under dynamic conditions.

Implemented sensor calibration and compensation algorithms for bias correction, alignment, and magnetic disturbance mitigation.

Developed **Qt-based desktop** software for real-time data visualization, logging, monitoring, and calibration via direct hardware communication.

** More details and visual examples of my projects are available on my website:*

<https://sadegh-mehrabi.github.io/resume>